BAMAT207-Probability and Statistics
Tutorial-2

1. Suppose that a Central University has to form a committee of 5 members from a list of 20 candidates out of whom 12 are teachers and 8 are students. If the members of the committee are selected at random, what is the probability that the majority of the committee members are students?

2. 8 coins are tossed at a time, 256 times. Find the expected frequencies of successes (getting a head) and tabulate the results obtained.

3. 20% of the bolts produced by a machine are defective. Obtain the probability distribution of the number of defectives in a sample of 5 bolts chosen at random.

4. Four coins are tossed simultaneously. What is the probability of getting
   (i) 2 heads and 2 tails, (ii) at least two heads, and (iii) at least one head.

5. Six dice are thrown 729 times. How many times do you expect at least three dice to show a five or six ?

6. A machine produces an average of 20% defective bolts. A batch is accepted if a sample of 5 bolts taken from that batch contains no defective and rejected if the sample contains 3 or more defectives. In other cases, a second sample is taken. What is the probability that the second sample is required?

7. Between the hours of 2 and 4 P.M., the average number of phone calls per minute coming into the switch-board of a company is 2·5. Find the probability that during one particular minute there will be:
   (i) no phone call at all, (ii) exactly 3 calls, (iii) at least 2 calls.

8. A manufacturer, who produces medicine bottles, finds that 0·1% of the bottles are defective. Bottles are packed in boxes containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles. Using Poison distribution, find how many boxes will contain : (i) no defective, (ii) at least 2 defectives.

9. The distribution of typing mistakes committed by a typist is given below. Assuming a Poisson model, find the expected frequencies.

Mistakes per page:	0	1	2	3	4	5
No. of pages:	142	156	69	27	5	1

   If a random variable X follows Poisson distribution such that P(X = 1) = P(X = 2), find (i) the mean of the distribution (ii) P(X = 0) (iii) P(X > 2).

10. The discrete random variable X represents the number of times a biased coin must be tossed until 3 “heads” have been obtained. It is further given that Var(X) = 36.
    a) Calculate P (X = 9).

b) If the first "head" was obtained on the second toss, find P ($X \leq 9$) .

11. A discrete random variable X has negative binomial distribution, with E(X) =12 and Var (X) = 4. Determine the value of P (X = 12).

12. If the probability that a target is destroyed on any one shot is 0.5, what is the probability that it would be destroyed on 6th attempt?

13. A random number generator produces values uniformly distributed between 0 and 1. What is the probability that the value generated is greater than 0.75 or less than 0.25?

14. Let X be a normal variate with mean 30 and standard deviation 5. Find the probability that
(i) $26 \leq X \leq 40$
(ii) $X \geq 45$
(iii) $| X - 30 | > 5$.

15. A company finds that the time taken by one of its engineers to complete repair jobs has a normal distribution with mean 40 minutes and standard deviation 5 minutes. State what proportion of job takes
(i) less than 35 minutes (the company charges Rs. 20 if the job takes less than 35 minutes).
(ii) between 35 and 48 minutes (the company charges Rs. 40 if it takes between 35 and 48 minutes).
(iii) more than 48 minutes (the company charges Rs. 70 if it takes more than 48 minutes).

16. In a normal distribution, 31% of the items are under 45 and 64% are over 64. Find the mean and variance of the distribution.

17. The number of kilometers that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 km. If the owner desires to take a 5000 km trip, what is the probability that he will be able to complete his trip without having to replace the car battery? Assume that the car has been used for the same time.

18. The amount of time that a watch will run without having to be reset is a random variable having an exponential distribution with mean 120 days. Find the probability that such a watch will
(i) have to be reset in at least 180 days, and
(ii) not have to be reset in at least 180 days.

19. The time in hours required to repair a machine is exponentially distributed with parameter $\lambda = 1$.
(a) What is the probability that the repair time exceeds 2λ?
(b) What is the probability that a repair takes at least 100 hours given that its duration exceeds 9λ?

20. The mileage which car owners get with a certain kind of radial tire is a random variable having an exponential distribution with mean 40,000 km. Find the probabilities that one of these tires will last
(i) at least 20,000 km, and
(ii) at most 30,000 km.

21. A passenger arrives at a local railway platform at 10 a.m. Knowing that the local train will arrive at some time uniformly distributed between 10 a.m. and 10:30 a.m., what is the probability that he will have to wait at least 10 additional minutes?

22. If X is uniformly distributed in the interval $(-3,3)$, find
    (a) $P(X > 2)$, $P(|\ X\ | < 2)$, $P(|\ X - 2\ | < 2)$, $P(|\ X - 2\ | > 2)$.
    (b) Find K such that $P(X > K) = \frac{1}{3}$.

23. The weights of packages are normally distributed with mean 5.2 kg and standard deviation 0.3 kg. If a sample of 36 packages is selected, what is the probability that the sample mean exceeds 5.3 kg?

24. The heights of students are normally distributed with mean 68 inches and standard deviation 3 inches. What percentage of students are taller than 72 inches?

25. The waiting time for a customer service call follows a gamma distribution with $\alpha = 3$ and $\beta = 2$ (in minutes). Find the probability that a customer waits more than 10 minutes.

26. The lifetime (in thousands of hours) of a component follows a Weibull distribution with a = 0.01 and b = 2. Find the probability that the component lasts more than 10,000 hours.

27. For a Weibull distribution with a = 0.01 and b = 2, find P(X > 10).

28. For a t distribution with $\nu = 15$, find the value of t such that P(T > t) = 0.025.

29. An electrical firm manufactures light bulbs that have a life normally distributed with mean 800 hours. Find (1) the probability that a bulb burns more than 778 hours and (2) the probability that a bulb burns between 778 and 834 hours.

30. The waiting time at a help desk follows an exponential distribution with mean 5 minutes. What is the probability that a customer waits less than 3 minutes?

31. In an examination, a student passes if he secures 30% or more marks. He is placed in the first, second or third division accordingly, as he secures 60% or more marks, between 45% and 60% marks and marks between 30% and 45% respectively. He gets a distinction in case he secures 80% or more marks. It is noticed from the results that 10% of the students failed in the examination, whereas 5% of them obtained distinction. Calculate the percentage of students placed in the second division. (Assume that marks are distributed normally).

32. Buses arrive at a specified bus stop at 15 minutes intervals starting at 7 a.m. that is 7 a.m., 7.15 a.m., 7.30 a.m., etc. If a passenger arrives at the bus stop at a random time which is uniformly distributed between 7 and 7.30 a.m. find the probability that he waits

    (a) less than 5 minutes
    (b) at least 12 minutes for a bus.

33. The time in hours required to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{2}$.
    (i) What is the probability that the repair time exceeds 2 hours?
    (ii) What is the conditional probability that a repair takes at least 10 hours, given that its duration exceeds 9 hours?